

CS23532 – COMPUTER NETWORKS – LAB MANUAL

Practical – 4

AIM

To set up and configure a Local Area Network (LAN) using a network switch and Ethernet cables, assign IP addresses to the connected computers, and verify connectivity and file sharing between the systems.

DESCRIPTION

A **Local Area Network (LAN)** is a network that connects computers and other devices within a limited geographical area such as a laboratory, office, school, or home. A LAN allows connected devices to communicate with each other and share resources such as files, printers, and network services.

In this experiment, four computers are connected to an **8/16/24-port network switch** using Ethernet cables. Each computer is assigned a unique IP address belonging to the same network. The switch forwards data between the connected computers, allowing them to communicate with each other.

REQUIREMENTS

- 4 Computers
- 1 Network Switch
- 4 Ethernet Cables
- Network interface ports
- Command Prompt

PROCEDURE

- 1 Plan the LAN topology and place the four computers and network switch appropriately.
- 2 Connect each computer to the network switch using an Ethernet cable. Connect one end of the cable to the computer's Ethernet port and the other end to a port on the switch.
- 3 Configure the IPv4 address of each computer through **Network Settings** → **Ethernet/Local Area Connection** → **Properties** → **Internet Protocol Version 4 (TCP/IPv4)**.
- 4 Assign the following IP addresses with subnet mask **255.0.0.0**:
PC1 – 10.1.1.1
PC2 – 10.1.1.2
PC3 – 10.1.1.3
PC4 – 10.1.1.4
- 5 Configure the network switch and assign the switch management IP address as **10.1.1.5** with subnet mask **255.0.0.0**.
- 6 Check connectivity by opening Command Prompt on each computer and using the **ping** command. For example: **ping 10.1.1.2**
- 7 Select a folder on one computer, open its **Properties**, select the **Sharing** tab, and share the folder with users on the same LAN.
- 8 From the other computers, access the shared folder and verify that files can be accessed through the LAN.

RESULT

Thus, a Local Area Network was successfully established using four computers, a network switch, and Ethernet cables. IP addresses were assigned to all devices, connectivity was verified using the **ping** command, and file sharing between the connected computers was successfully demonstrated.